

Use Case Flow

Gyaan S Shetty pes1ug24cs178

Problem Statement 49 : Webhook Ingestion and Retry Mechanism Hub

Goal	Ingest and Deliver Webhook
Actor	Third Party Service, with help from System Operator
Main flow	1. A Third Party Service sends a webhook to the ingestion endpoint. 2. The system verifies the HMAC signature of the request 3. The system buffers the webhook and writes a log row (source, time, status = RECEIVED). 4. The system forwards the payload to the configured target endpoint. 5. The target endpoint replies with a success response (http 2xx). 6. The system updates the log row to status = DELIVERED. 7. The system keeps the event in history for later auditing. 8. Ingestion process is completed successfully.
Alternae Flow	Delivery Failed At Step 5, if the target endpoint returns an error or times out : 5a. The system retries the delivery using exponential backoff (1s,2s,4s,8s). 5b. If a retry works, the flow continues from Step 6. 5c. If all 4 tries fail, the system moves the webhook to the Dead Letter Queue and notifies the System Operator.